

Repeated games

Will it end?



Podium:

	High	Low
High	3	0
Low	5	1

Low is a dominant action in the one-shot game, which is exactly the problem.

Your payoff as the row bidder.

Played once, what should you do?

Played once, what do you both get?

And if you had both bid High?

We stop on a 1, so the game carries on with probability $\delta =$.

The expected number of rounds is $\frac{1}{1 - \delta} =$.

Now we carry on *only* for a 5 or a 6, so $\delta =$ and the expected number of rounds is .

Did the shorter game change how you played?

Grudger plays High until you play Low, then Low for ever. Hold your opponent to it.

You conform, and play High for ever

$$3 + 3\delta + 3\delta^2 + \dots =$$

You deviate to Low in round one

$$5 + \delta + \delta^2 + \dots =$$

So cooperating is worth it when

 $\geq$

$$\frac{3}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \iff \delta \geq \boxed{}$$

Our first die gives $\delta = 5/6$, so cooperation be sustained. The second gives $\delta = 1/3$, so it .

Which side of the line did you guess before we played?

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In a round you know is the last, the future is worth , so you should bid .

Both of you know that, so in the round before it the future is also worth nothing, and you bid . Keep going backwards and cooperation _____.

So the die was doing real work: not by making the game long, but by making the end _____.

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What we just did in the room is **Question 1** on the Repeated Games page.

- (a) Interpret $\delta = 5/6$ and give the expected number of rounds. [2]
- (b) Discounted payoffs against Grudger, conforming and deviating. [6]
- (c) Is cooperation sustained at $\delta = 5/6$? [4]
- (d) The smallest δ that sustains it, and what our die meant. [5]
- (e) The same for a forgiving trigger that punishes for two rounds only. [7]

vknight.org/gt/topics/repeated-games.html